Voice-coil driven flapping-wing micro-air-vehicles (FWMAVs)

Uncrewed aerial vehicles (UAVs) / TRA370

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Contents

1 Introduction

The way insects fly is among one of the most astonishing evolutionary achievements. The efficiency and extremely high maneuverability of these creatures have long sparked the ambition to recreate such systems, dating back to the very beginning of aviation.
But recreation of such flight mechanisms poses significant challanges, due too the small size and highly complex mechanisms necessary to implement full control in all degrees of freedom.
Here at Chalmers several insectoid micro air vehicle prototypes (which will be referred to as MAV) have been developed, the latest alteration being a voice coil driven MAV.

2 Motivation

At first glance it might seem counterintuitive to strive for the recreation of these highly complex mechanisms, as mankind already developed unmanned multi-rotor aerial vehicles which provide high maneuverability and can be manufactured at very low costs. However, these systems have limitations regarding how small they can be constructed and are also not a very energy efficient form of flight.
Insectoid flight mechanisms on the other hand can provide a comparatively energy efficient form of flight at small scales and are therefore desirable to investigate further.

3 Objectives

The main objective of this project was rather simple: Produce as much lift as possible with the provided prototype.
This might seem easy at first, but given the complexity of such systems, it is not trivial. To generate as much lift as possible, all aspects of this system had to be taken into account. The main working areas where the choice of the input signal, understanding the underlying mechanical concepts and measuring the output of the system in a way that all necessary information can be gathered.

4 Development of the setup

In the beginning we were provided with a voice coil actuated MAV prototype and different pairs of wings, which can be seen in figures 1 and 2. Nearly all mechanical parts of the prototype, from the body frame too the wing hinges, have been manufactured using stereolithography. This technique allows for the required tolerances to be met and results in a very precisely manufactured final product, which was necessary in this case due to the small scale of the parts.

Figure 1: render of the provided MAV prototype

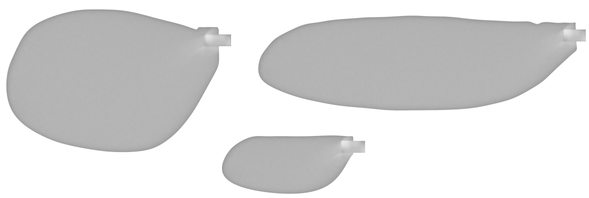

Figure 2: render of the provided wings

4.1 Scaffolding structure

To provide free movement of the wings and at the same time hold the prototype in place during test runs, the first thing we did was manufacturing a simple scaffold structure using filament 3D printing, as for these parts high precision of the end product was not a major concern.

4.2 Development of electrical circuit and control software

After the rather simple fist step of constructing the scaffold, the next task was the development of the electrical circuit.
We rather fast came too the conclusion, that it would be beneficial to be able to alter

the generated output signal, therefore we decided to go for a setup including a micro controller which controls a motor driver. The motor drivers output signal is a classic PMW signal.
Due to availability in the lab and ease of use in prototyping, we used an Arduino Uno as micro controller. The process of choosing a suitable motor driver required a little bit more thought. We started out with looking at the voice coil data sheet and calculating the maximum voltage and current the motor driver must be able to supply, visible in table 1. After knowing the demands for the motor driver, we decided to use the low cost standard AZ-L298N Motor Driver Board, visible in figure 3.

Duty	**W**	**A**	**V**
10%	16	2.4	6.67
20%	14.44	2.22	6.50
30%	12.89	2.04	6.30
40%	11.33	1.87	6.07
50%	9.78	1.69	5.79
60%	8.22	1.51	5.44
70%	6.67	1.33	5.00
80%	5.11	1.16	4.42
90%	3.56	0.98	3.64
100%	2.00	0.80	2.50

Table 1: maximum Power (W), Current (A), and Voltage (V) vs. Duty Cycle of the voice coil

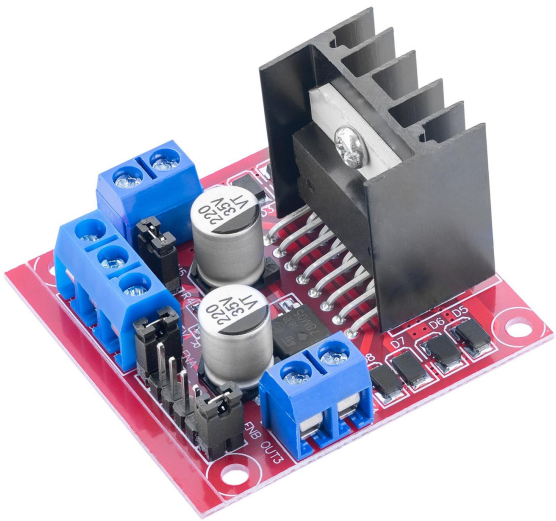

Figure 3: motor driver

The further work done to connect the components was trivial and therefore will not be discussed in further detail. After all components where connected, LEDs were inserted as load on the motor driver output side to have a visual feedback, which proved quite useful in the first stages of the control software development.
After we understood how to properly control the motor driver, the next step was to think about the driving signal for the voice coil we wanted to use. Many ideas came up, like to use a regular PWM signal, a linearly increasing or decreasing signal, a sinus curve and many more. In the end we came to the conclusion, that in our case a rather simple rectangular signal would be the best choice. To maximize the power provided too the voice coils when changing direction of movement at the beginning of a half cycle and minimize the power needed to turn around at the end of a half cycle, the generated signal is set to 100% duty at the beginning of a half cycle and is set to 0% at the end. Basically, the driving signal is a PWM signal, but there is only one high and one low phase per half cycle. The final signal form can be seen in figures 4 and 5, the different curves will be discussed later in more detail.

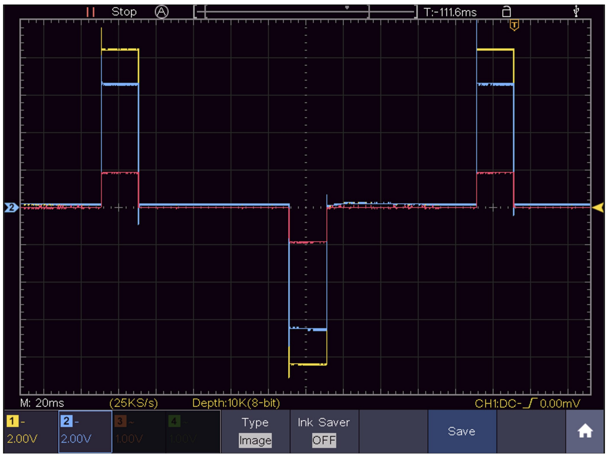

Figure 4: driving signal 20% duty cycle

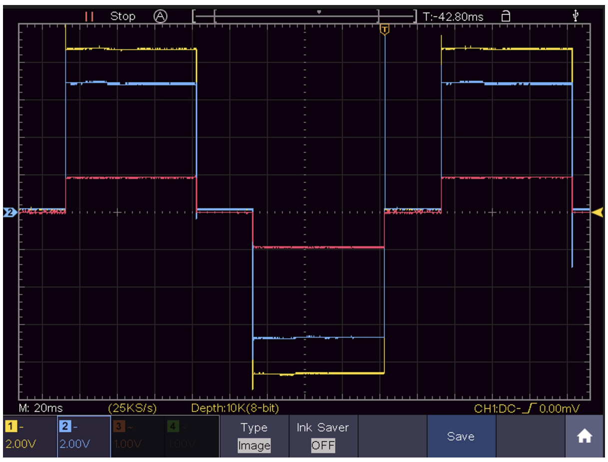

Figure 5: driving signal 70% duty cycle

4.3 Choice and setup of the measurement devices

There where multiple parameters of interest we wanted to gather during our test runs, namely the produced lift, the consumed power, input signal parameters and details on the wing movement.
To measure the produced lift, we used the provided lab scale and placed the prototype upon it. When actuated, the MAV produces lift and therefore a negative weight can be seen on the scale, when calibrated correctly.
To measure the consumed power, we needed to gather information about the correct voltage over and current through the voice coil. To do that, we inserted a 1Ω resistor in series too the coil in the circuit to measure the current. Using an oscilloscope and the differential measuring technic, we where able to gather the needed information. In figures 4 and 5 the yellow channel (CH1) shows the voltage over both resistor and coil, the blue channel (CH2) shows the voltage over the coil, both in reference too the ground potential. By using the math function to subtract CH2 from CH1 it is possible to determine the voltage over the resistor (red graph). As we used a 1Ω resistor, the measured voltage is equal to the current through the serial circuit.
As we control the input signal with our setup and also verified its accuracy through oscilloscope measurements, we already know the input signal parameters of interest, namely frequency, amplitude and duty cycle.
To gather information about the wing movement, we used an Apple iPhone 15 Pro to make slow motion videos of the MAV. For the phone to be in a stable position, we used a small vice and placed the phone within it.

4.4 Different wing hinge alterations

As the experiments continued, we decided to manufacture additional wing hinges. The variable of change in the construction was the stopping angle, which de facto determines the angle of attack. Again we used stereolithography, as high precision was required. The different hinge stoppers can be seen in figure 6.

Figure 6: hinge stoppers (left too right): 75°, 65°, 55°, 45°, 35°, 25°, 15° wing rotation

5 Lab test runs

5.1 introduction

As we were able to control our setup in the desired ways and also gather measurement data, the next step was to think about which test runs we wanted to conduct. Our input signal provided us with three variables: amplitude, frequency and duty cycle. We decided to start with the amplitude alteration experiments, after that we looked into the effects of different frequencies and duty cycles on different wings and at the end we ran some test with the altered wing hinge stoppers.

5.2 Amplitude alteration

For this test run we held the duty cycle constant and changed the input voltage in 1 V steps from 5 to 14 V. We conducted different runs in 1 Hz steps from 3 too 8 Hz.
We expected a linear relationship between the amplitude of the input signal and the produced lift, as a higher amplitude would mean that the voice coil receives more power, which then should produce a higher force, which should result in more powerful flapping. The collected data supported our assumption (figure 7), with only the 3 and 8 Hz run not showing a linear relation ship. This was mainly due to mechanical reasons.

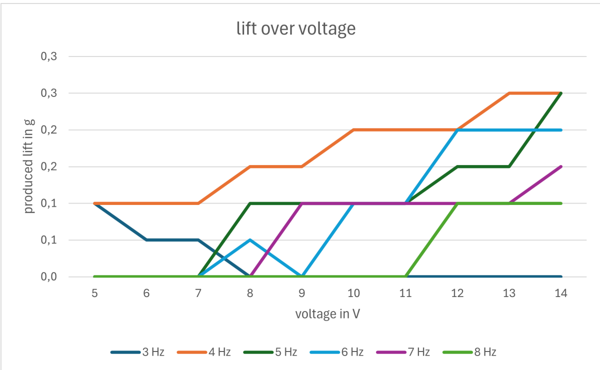

Figure 7: lift over input voltage for different frequencies

5.3 Duty cycle alteration

After the last test run, we lubricated the wing hinges with gear box oil, this significantly increase the produced lift. Therefore from now on, measurement data is not comparable anymore too the voltage test run. The linear relationship between voltage and lift output was assumed to be unchanged, therefore we selected the highest voltage (14 V), as this delivered the best results.
We made test runs altering the duty cycle in 10% steps, from 10 too 100% and repeated these experiments again for different wings in in 1 Hz steps from 3 too 8 Hz steps.
Figure 8 shows the best test runs for different wings.

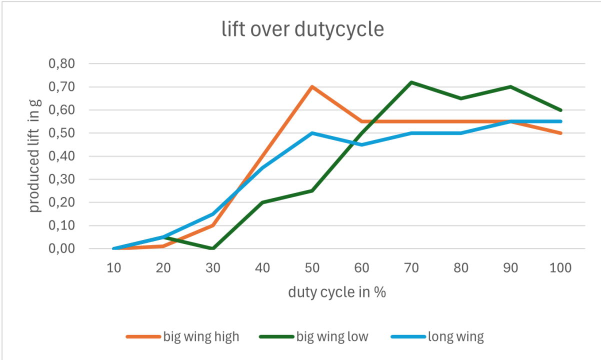

Figure 8: lift over duty cycle for different wings

The best test runs were conducted for the big wing mounted in the upper position at 7 Hz, 70% with 0,72 g of lift produced, for the big wing mounted in the lower position at 6 Hz, 50% with 0,7 g of lift produced and for the long wing at at 4 Hz, 70% with 0,5 g of lift produced.
In figure 8 it is clearly visible, that an increase of the duty cycle over 70% does not increase the performance anymore. This is most likely because a too long power phase hinders the voice coil from smoothly turning around at the stopping point.
For runs with the small wings was no time anymore.

5.4 Wing hinge test runs

As mentioned before, we where also interested in the effects of different wing hing stoppers. We assumed, that a lower maximum wing rotation angle will decrease lift output and hoped for an increase if the wing could rotate further.

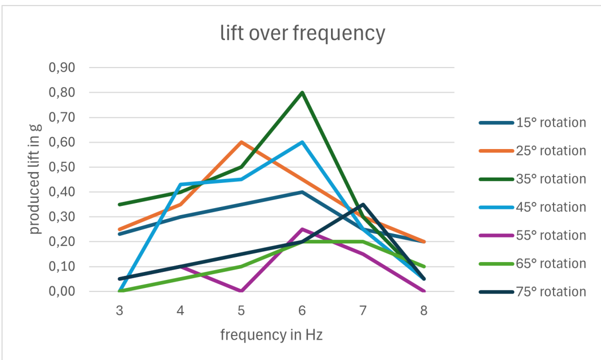

Figure 9: lift over frequency for different wing hinge stoppers

As can be seen in figure 9, our the experiment showed, that a smaller maximum wing rotation angle decreases lift output. From 15° too 35° the lift performance increased steadily, peaking at 35°. 45° provided good results too, but further increasing the angle drastically reduced the lift output. Therefore it is not recommended to use maximum wing rotation angles greater than 45°. This is most likely because the wings can rotate freely and are just actuated through their own inertia. When the maximum rotation angle gets too high, the wings can not reach their desired stopping point anymore, because it takes them too long to rotate in position. Through that the whole wing motion gets messed up and therefore lift output drops close too zero.

6 Conclusion for the electro-mechanical part of the project

The maximum lift achieved was only 0.8 g. This is not even close to the needed lift for take off (absolut minimum 11 g, which is the weight of the flapper without support structure). As we have been pushing the voice coil to its limits (it got quite hot, especially during the high duty cycle runs), drastic improvement without changing the prototypes actuator seems unlikely.
Another area of improvement would be the wings, but designing and manufacturing new wings is not a trivial task. Also the underlying mechanical structure could be altered, one could insert springs in the system to achieve a smoother and more energy efficient movement of the voice coil.
Also should be mentioned, that it is not possible with this prototype to achieve controlled flight, as for now only the lift force can be controlled, in terms of flying only up or down movement can be controlled.

7 Wing tracking system

7.1 Introduction

The wing tracking system aims to extract wing movement information using computer vision techniques. The idea is to make a recording of the wing movement and place colored markers on relevant spots of the wing. With the help of a flexible, extensible python framework for tracking colored markers, different evaluation cases should be easily configurable. The system is built with modularity in mind, allowing for easy adaptation to new test cases, marker configurations and analysis requirements.

7.2 Description

To analyze the motion patterns of the wing, LED markers are attached to relevant anatomical points depending on the evaluation case. The tip, the root, and the trailing edge for example are crucial to analyze the wing movement from the top-down perspective. Recordings of the flapping motion are captured and the corresponding videos are served as input for the tracking pipeline.

7.2.1 Histogram analysis

Prior to tracking, the region of interest is cropped and the HSV (Hue, Saturation, Value) color bounds for each marker are gathered using a histogram-based analysis tool. These bounds are then used to generate color masks that isolate the marker regions in the image. This process can be seen in figure 10.

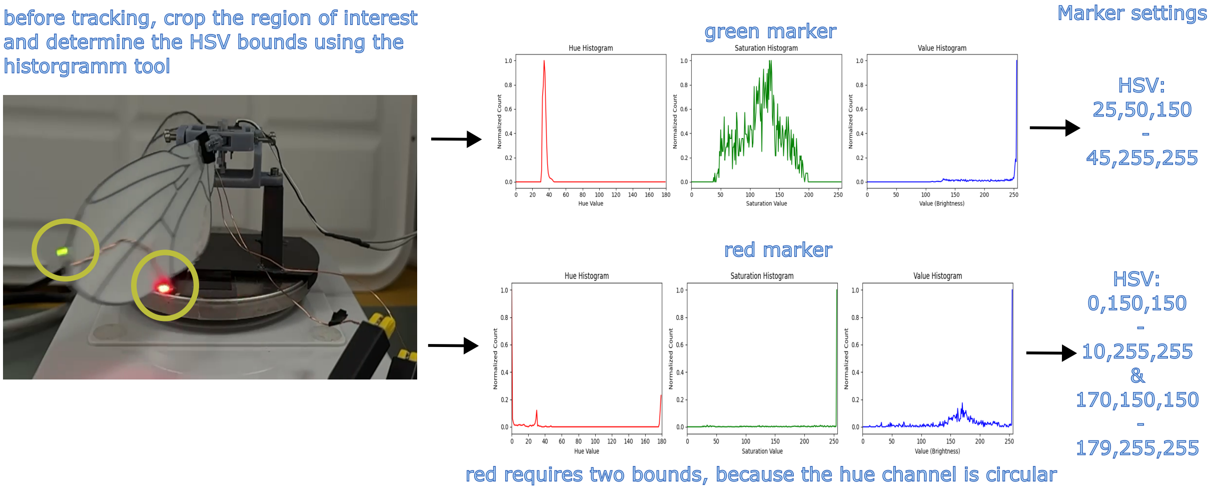

Figure 10: Extracting HSV-range of color markers based on Histogram from one frame

7.2.2 Tracking pipeline

Each video frame is individually processed to detect the marker positions. Once the masks are created, using the defined HSV bounds for each marker, a detection algorithm of choice is applied to identify marker positions within each frame. The markers are then visualized on the original frame, with historical marker positions plotted to reveal the trajectory over time, which can be seen in figure 11. Additionally, if an ArUco marker is being detected during the frame, the pixel distances will be transformed to real life values, in that case millimeters. The processed video, along with a log file of detected positions, is then exported after the final frame has been processeced.

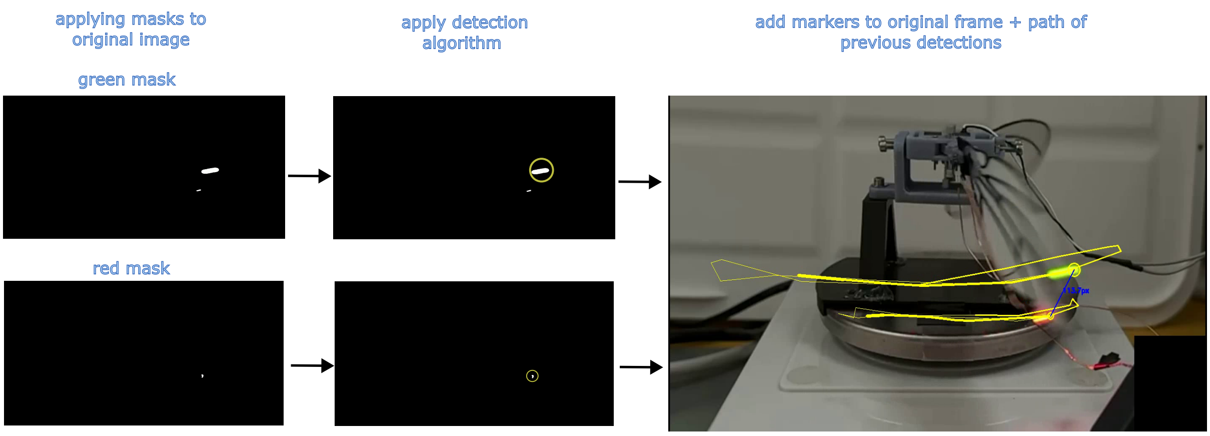

Figure 11: Application of tracking on one frame

7.3 Conclusion and future prospects

Several tracking algorithms were tested, each with specific advantages and trade-offs. Some algorithms offered the benefit of real-time processing, making them suitable for applications requiring immediate feedback. Others provided higher detection accuracy but at the cost of increased computational load, making them more suitable for offline analysis.

For optimal tracking performance, several environmental and configuration factors must be considered. It is crucial to ensure that all markers remain visible throughout the recording session. The recording environment should avoid using background colors that fall within the HSV bounds of the markers, as this could result in false positives. Additionally, even lighting conditions and a high frame rate are essential to improve the accuracy and temporal resolution of the tracking system.

Further improvements to the system would be:

- **Generalized marker specifications:** The number and type of markers (e.g., root tip, trail edge) are hardcoded in each tracking mode. A major future goal is to make the system more generic by allowing the user to specify the number and type of markers required for a test when initializing.
- **Advanced calculations and analysis:** Calculations for wingbeat frequency, amplitude, or other biomechanical parameters are not yet implemented and would be a good addition to the current framework. This will most likely require to make use of all frames and not only focusing on individual frames as it stands currently.
- **3D analysis and camera calibration:** The current version assumes a 2D projection. Combining intrinsic/extrinsic camera parameters with ArUco marker detection would allow for a 3D marker detection and enable more sophisticated biomechanical studies